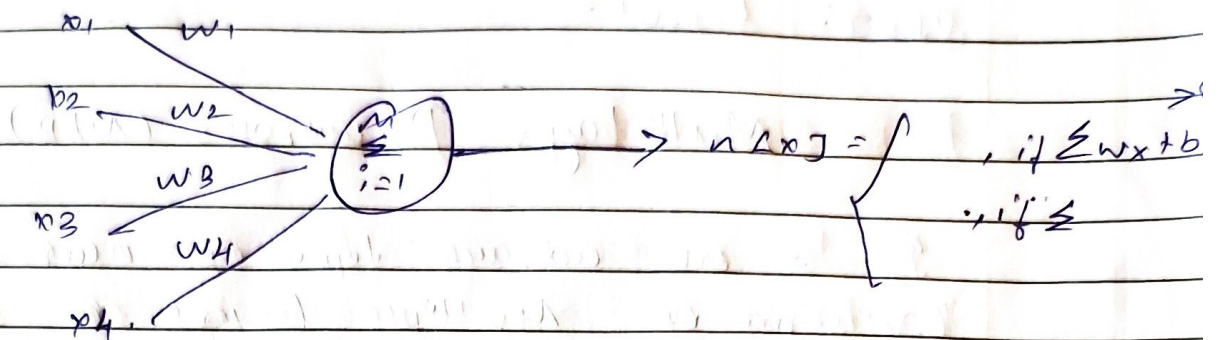


Perceptrons :-

A perceptron receives one or more input values, multiplies each input by a corresponding weight, adds a bias and applies an activation function to produce an output.



Structure of perceptrons.

1. Input layer
2. Weights
3. bias
4. Summation unit
5. Activation function

Steps:

- Step 1: Initialize weights.
- Step 2: Provide training data.
- Step 3: Compute the net input.

$$Z = \sum_{i=1}^n w_i x_i + b$$

Step 4: Apply the Activation function.

Step 5: Compare with the target output.

Step 6: Update weight.

$$W_{new} = W_{old} + \eta (t - y) x.$$

Similarly update the bias.

$$b_{new} = b_{old} + \eta (t - y).$$

Step 7: Repeat.

Multi Layer Perceptron (MLP)

It is an advance type of neural network consisting of multiple layers of neurons including one or more hidden layers.

Q.1

Input: $x_1 = 2$, $x_2 = 3$.

Weight: $w_1 = 0.5$, $w_2 = 0.4$.

Bias $b = -1$

$$\begin{aligned} Net &= x_1 w_1 + x_2 w_2 + b \\ &= (2)(0.5) + (3)(0.4) - 1 \\ &= 1.2 \end{aligned}$$

$Net \geq 0$.

Ans: 1.2.

Now update the values.

Learning weight = 0.2

Target = 1

Output = 0

Input = 3. Old weight = 0.4.

Formula:

$$W_{new} = W_{old} + \eta (\text{Target} - \text{output})$$

$$= 0.4 + 0.2(1-0)(3)$$

$$= 0.4 + 0.6$$

$$= 1.0$$

Updating bias:

$$\text{Old bias} : 0.5$$

$$\text{Learning rate} = 0.1$$

$$\text{Target} = 0$$

$$\text{Output} = 1$$

Solution:

$$b_{new} = b + \eta (\text{Target} - \text{output})$$

$$= 0.5 + 0.1(0-1)$$

$$= 0.4$$

Curse of Dimensionality

It refers to the challenges that arise when working with high-dimensional data where increasing the no. of features leads to reduced model performance and higher computational cost.

It becomes more difficult to identify meaningful patterns.

How to overcome :-

1. Feature Selection.
2. Use Appropriate algorithm.
3. Dimensionality Reduction.

Common techniques.

PCA principal Component Analysis.

Autoencoders in Deep learning Learning (AE).

Probability:

It is the measure of how likely an event is to occur. The probability of an event lies between 0 and 1.

Probability $\Rightarrow 0 \rightarrow$ Impossible event.
probability $\Rightarrow 1 \rightarrow$.

$$P(E) = \frac{\text{No. of Outcome}}{\text{Tot.}}$$

i) Addition, if ~~mutually~~

$$P(A \cup B) = P(A) + P(B)$$

ii, Multiplication.

$$P(A \cap B) = P(A) \times P(B)$$

iii, Complement Rule.

$$P(A') = 1 - P(A)$$

3. Random Variable:-

a) Discrete random variable:-

Example:-

No. of defective bulbs in a box of 100.

Possible values:-

$$x = 0, 1, 2, 3, \dots$$

b) Continuous random variable:-

4) Probability Distribution:-

is describes how probabilities are distributed

5) Mean (Average)

Average values of all observations.

$$\text{Mean} = \frac{\sum x}{n}$$

6) Variance:-

Measures how far the data values are spread from the Mean.

A small variance mean value are close to the mean, while a large variance indicates they are widely spread.

$$\text{Var}(x) = E(x^2) - [E(x)]^2$$



Standard deviation:

is the square root of the variance. It indicates how much the data values deviate from the mean.

$$S = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

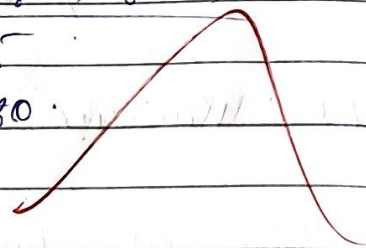
Formula ~~is~~ =

Real time ML Example:-

Student	Predicted Marks
A	70
B	75
C	80
D	85
E	90

$$\text{Mean} = \frac{70 + 75 + 80 + 85 + 90}{5} = 80$$

Avg predicted mark is 80.



Supervised learning Algorithm.

is a type of Machine learning algorithm in which a model learn a labeled training data.

Types:

Regression :-

is a supervised learning technique used to predict a continuous numerical output based on one or more input variables.

Example: House price prediction.

$$y = mx + c$$

Classification:

Used to predict a categorical output or class label.

Types:

i) Binary classification.

ii) Multi-class classification.

Decision Tree:-

Definition:-

is a supervised learning algorithm used as a tree-like structure to make prediction.

Root node:

Decision / Internal node

Branch

Leaf node.

K-Nearest Neighbors (KNN)

Predict class of New data points
on the basis of its k-nearest neighbors

Euclidean Distance formula:-

$$A(x_1, y_1) \\ B(x_2, y_2)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Class A : A(1,1)

B : B(2,2)

C : C(6,6)

new point is

P(2,1)

Let, $k=3$ find the predicted class.

Step 1:- Calculate distance from P to point A.

$$P = (2,1), A = (1,1)$$

$$d = \sqrt{(2-1)^2 + (1-1)^2} = \sqrt{1}$$

Step 2: P to point 2.

$$P(2,1) \quad B = (2,2)$$

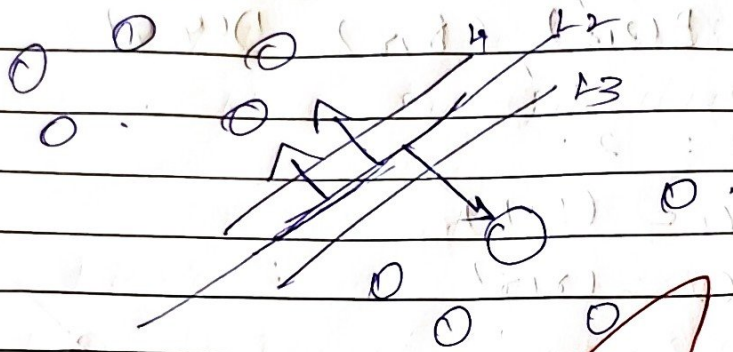
$$d = \sqrt{(2-2)^2 + (1-2)^2} = \sqrt{1}$$

Step 3:

$$d = \sqrt{(2-6)^2 + (1-6)^2} = \sqrt{16+25} = \sqrt{41} \approx 6.4$$

SVM: Support vector machine.

mainly used for classification, although it can be used for regression.



data points which is closest to the decision boundary is called support vectors

Unsupervised learning:-

is a type of ML in which the algorithm learns patterns and structure from unlabeled data.

1. K Means clustering.
2. Hierarchical clustering.

1. K mean:-

K — No. of cluster.

Centroid — center of cluster.

Cluster - Group of similar data points.

Distance - Euclidean distance.

$$\text{and, } \begin{matrix} A(x_1, y_1) \\ B(x_2, y_2) \end{matrix}$$

the distance is

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$A(1,1) \quad B(2,2) \quad C(8,8) \quad D(9,9)$$

$$K = 2$$

$$C_1 = (1,1)$$

$$C_2 = (8,8)$$

Step 1:

$$C_1 = C_1(1) : A(1,1)$$

$$d(A, C_1) = \sqrt{(1-1)^2 + (1-1)^2}$$

$$\text{cluster 1} = 0$$

$$C_2 = C_2(1) : A(1,1)$$

$$d(A, C_2) = \sqrt{(8-1)^2 + (8-1)^2}$$

$$= \sqrt{49 + 49}$$

$$\text{cluster 2} = \sqrt{98} = 9.90$$

$\therefore A$ belongs to cluster 1.

Hierarchical Clustering:

Tohmanu first creates a hierarchy of clusters.

1. Agglomerative clustering.

Each data point = One cluster.

2. Divisive clustering

All data points = One cluster.

Ex:

$A=1$, $B=2$, $C=8$, $D=9$.

Step 1:

$$d(A, B) = |1 - 2| = 1$$

$$d(A, C) = |1 - 8| = 7$$

$$d(A, D) = |1 - 9| = 8$$

$$d(B, C) =$$

$$d(B, D) = |2 - 9| = 7$$

$$d(C, D) = |8 - 9| = 1$$

Step 2:

$$d(A, B) = 1$$

$$d(C, D) = 1$$

$\{A, B\}$

$\{C, D\}$

Representation Learning

is the process of automatically discovering and extracting meaning from features.

Types:

1. Low Level Features.

Edges

Corners

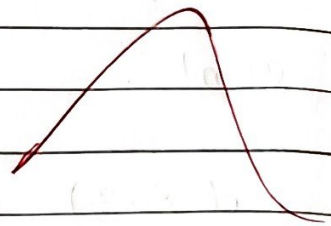
Lines

2. Mid Level features.

Eyes

Nose

Doors.



3. High level

Human face
car

Neural Network :-

is a computational method which is interconnected neurons :-

Biological Inspiration :-

Human Brain.

Neuron

Dendrites

Axon.

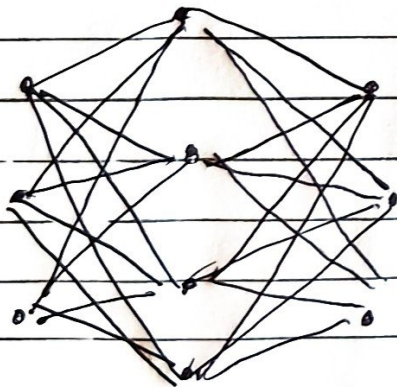
Synapse

3 imp Concepts:

- i) width & depth
- ii) model capacity.
- iii) Network Architecture.

Deep Neural Network :-

Neural Network :-



Regulus